INCH-POUND
A-A-59988
March 23, 2018

COMMERCIAL ITEM DESCRIPTION

TRUCK, FIRE FIGHTING (AIRCRAFT RESCUE AND FIRE FIGHTING) (ARFF), AIR TRANSPORTABLE

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This CID covers the 1,000 gallon version of the commercially available diesel engine driven ARFF. An ARFF vehicle is intended to carry rescue and fire fighting equipment for rescuing occupants and combating fires in aircraft.

2. SALIENT CHARACTERISTICS. The ARFF vehicle shall be built in accordance with the applicable requirements of National Fire Protection Association (NFPA) 414, Current Edition at the time of procurement.

2.1 Administration.

2.1.1 Manuals.

2.1.1.1 Technical manuals. The overall format for the manuals may be military specification, commercial specification, or a combination of both. Each technical manual shall have a title page. One set of engine and transmission parts, service, and operator's manuals shall be packed with each vehicle. An additional two complete sets of engine and transmission parts, service, and operator's manuals shall be sent to the procuring activity for stock.

The contractor shall submit digitized technical manuals and printed paper copies. The contractor may recommend the delivery of files developed in a "native" format as the final delivery medium. Native format includes MS Word, Framemaker, or Interleaf. However, all recommendations for native formatted data or graphics shall be approved by the Technical Order Manager prior to any work being performed.

Beneficial comments, recommendations, additions, deletions, clarifications and any data that may improve this document should be sent to: U.S. AIR FORCE, AFLCMC/WNZEB, 295 Byron Street, Robins AFB, GA 31098 or emailed to SPEC99@us.af.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/ .

AMSC N/A FSC 4210

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a. The contractor shall validate the technical manuals for accuracy prior to submission to the procuring activity for verification. The contractor shall submit one complete set to the procuring activity for verification 30 days before the first production inspection. Any changes or corrections noted by the procuring activity shall be corrected and updated pages or manuals shall be submitted to the procuring activity.

b. Once approved by the procuring activity and a Technical Order (TO) number has been assigned, the contractor shall pack two complete sets of both hard copy and CDs with each vehicle. All manuals provided must be vehicle specific and reflect the configuration of the vehicle as built. Generic manuals are not acceptable. One complete set of manuals in CD format shall be sent to the procuring activity for stock.

c. The contractor shall grant the United States Air Force a non-exclusive, non-assignable, royalty free U.S. Government (Government) Purpose License, to scan into CENTRA (the USAF repository) and to reproduce and distribute (either electronically or via hard medium) copies or facsimiles of manuals produced and distributed by the contractor for this CID. These rights extend to Government agencies only, and the data contained in the technical manuals are not to be sold, disclosed or otherwise provided to any other entity or entities outside the Government. The license shall remain in effect as long as the vehicles described in the required technical manuals remain under Government control and usage. In addition, the contractor shall grant permission for the Government to place an Air Force TO number and publication date as well as distribution, warning, handling, and destruction statements on the cover / title pages of the TO.

2.1.1.1.1 Operator's manual. The operator's manual shall include all information required for the safe and efficient operation of the vehicle, including the fire extinguishing equipment, and any special attachments or auxiliary equipment. The operator's manual shall include the following items at a minimum:

a. Location and function of all controls and instruments shall be illustrated and fully described.

b. Safety information that is consistent with the safety standards established by the Occupational Safety and Health Administration (OSHA).

c. Checks and adjustments in preparation for placing the vehicle in service upon receipt from the contractor.

d. Preparation for shipment or storage.

e. Warranty information and period of the warranty for the complete vehicle and for any component warranty that exceeds the warranty of the complete vehicle. Addresses and telephone numbers shall be provided for all warranty providers.

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f. General description of and step-by-step instructions for the operation of the vehicle and its fire extinguishing systems and auxiliary equipment.

g. Description of the post-operational procedures (such as draining or flushing).

h. Checklists for the daily maintenance inspection and mission readiness checks that the operator is expected to perform, including basic troubleshooting procedures.

i. Procedures for towing a disabled vehicle.

j. Procedures for changing a tire.

k. Schedules for required preventative maintenance and required periodic maintenance.

l. A line art drawing of the vehicle, including front, rear, left, and right side views, showing basic dimensions and weights (total vehicle and individual axle weights for the unloaded and fully loaded vehicle).

m. Step by step instructions for the foam testing system in accordance to NFPA 412 and NFPA 414 requirements.

2.1.1.1.2 Service manual. The service manual shall identify any special tools and test equipment required and shall cover troubleshooting and maintenance as well as minor and major repair procedures. The text shall contain performance specifications, tolerances, and fluid capacities; current, voltage, and resistance data; test procedures; and such illustrations and exploded views as may be required to permit proper maintenance by qualified mechanics. The manual shall contain an alphabetical subject index as well as a table of contents. The service manual shall contain at least the following, where applicable:

a. Fire fighting system schematic.

b. Hydraulic schematic.

c. Pneumatic schematic.

d. Electrical schematic.

e. Winterization schematic.

f. Fuel schematic.

g. Schedules for required preventative maintenance and required periodic maintenance.

h. Location, procedure, and interval for parts of the truck and equipment that require lubrication.

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2.1.1.1.3 Parts manual. The parts manual shall include illustrations and exploded views, as needed, to properly identify all parts, assemblies, subassemblies, and special equipment. All components of assemblies shown in illustrations or exploded views shall be identified by reference numbers that correspond to the reference numbers in the parts lists. All purchased parts shall be cross-referenced with the original manufacturer's name and part number. The parts identification manual shall provide the description and quantity of each item used per vehicle. The size, thread dimensions, torque specifications, and special characteristics shall be provided for all nonstandard nuts, bolts, screws, washers, grease fittings, and similar items. The manual shall contain a numerical index. The parts manual shall contain a list of all of the component vendor names, addresses, and telephone numbers referenced in the parts list.

2.1.2 Painting, plating, and corrosion control.

2.1.2.1 Finish. Exterior surfaces shall be prepared, primed, and painted with polyurethane paint in accordance with all of the paint manufacturer's instructions and recommendations. The exterior finish color shall be Desert Sand, Color Number 30313 of SAE AMS-STD-595**.** The interior of all compartments shall be painted with an impact resistant, textured coating that resists stains, scuffs, chips, and scratches. All exterior surfaces, including all normally bright metal and anodized parts and any interior surfaces visible with any compartment door open (but not the interior of the cab), shall be painted body color. This includes compartment shelves and mounting hardware, but does not include items mounted in the compartments. Non-metallic materials may be black or gray.

2.1.2.2 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotively compatible abutting surfaces is acceptable. The use of dissimilar metals separated by suitable insulating material is permitted, except in systems where bridging of insulation materials by an electrically conductive fluid can occur. Sealants or gel type gaskets materials shall be used between faying surfaces and butt joints.

2.1.2.3 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during service shall not be used or shall have a means of protection against such deterioration that does not prevent compliance with performance requirements. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Coatings shall be free from runs, sags, orange peel, or other defects.

2.1.2.4 Sound suppression material. Sound absorbing and suppression material shall be of the non-moisture absorbing type, so that condensation and rain water are not absorbed and held adjacent to metal structures and panels. Fiberglass, felt, and open cell foam shall not be used for sound suppression.

2.1.2.5 Exclusion of water. The design of the vehicle shall preclude water from leaking into, or being driven into, any part of the vehicle’s interior when stationary or moving. All windows, including doors, panels, covers, shall be provided with sealing arrangements to minimize water entry when the items are properly closed. Particular care shall be taken to prevent wetting of equipment, heat, and sound proofing materials. Sealed floors with drainage shall be provided for storage compartments, engine compartments, and other areas in the design that could collect and retain water.

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2.1.2.5.1 Fluid traps and faying surfaces. There shall be no fluid traps on any component. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

2.1.2.5.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

2.1.2.5.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall be sufficient to permit drainage of all fluids when the unit is in a 10° incline in any plane. The minimum size of each drain hole shall be 0.375-inch.

2.1.2.6 Reflective stripes. Vehicles shall be marked with white reflective striping to comply with the surface area coverage as prescribed in 4.2.4.1 of the NFPA Standard 414 for Aircraft Rescue and Fire Fighting Vehicles. Bright metal trim or anodized parts may interrupt the reflective stripes. The pattern shall be 10-inches wide with three reflective stripes (1-inch reflective, 1-inch body color, 6-inches reflective, 1-inch body color, and 1-inch reflective).

2.1.3 Identification plate. A permanently marked identification plate shall be securely mounted at the driver's compartment. The identification plate shall contain the following information:

a. NOMENCLATURE
b. MANUFACTURER'S MAKE AND MODEL
c. MANUFACTURER'S SERIAL NUMBER
d. GOVERNMENT REGISTRATION NUMBER
e. NATIONAL STOCK NUMBER (NSN)
f. VEHICLE CURB WEIGHT: lb (kg)
g. PAYLOAD, MAXIMUM: lb (kg)
h. GROSS VEHICLE WEIGHT (GVW): lb (kg)
i. FUEL CAPACITY AND TYPE: gal (ltr)
j. DATE OF MANUFACTURE: (month and year)
k. WARRANTY: (months and miles (km))
l. CONTRACT NUMBER
m. PAINT COLORS AND NUMBERS
n. LENGTH, WIDTH, AND HEIGHT OF VEHICLE IN INCHES AND CENTIMETERS
o. AXLE RATINGS: lb (kg)

2.1.3.1 Secondary identification plate. A second permanently marked information data plate shall be securely mounted on the interior of the driver's compartment door. The plate shall contain the information required by 4.2.5 of NFPA 414.

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2.1.3.2 Combined identification plate. A single plate that combines or contains the information required for both plates is acceptable.

2.1.4 Environmental conditions.

2.1.4.1 Temperature range. The vehicle shall be capable of satisfactory storage and operation in temperatures ranging from -40° to 125°F. The vehicle shall be equipped with a cab, chassis, and agent winterization system, permitting operation at -40°F. The winterization system shall not detract from the performance of the vehicle or the fire fighting system in ambient temperatures up to 125°F. The vehicle chassis winterization system shall maintain the engine coolant, lubricants, fuel, and electrical systems operational at ambient temperatures of -40°F. The vehicle agent winterization system shall provide sufficient insulation and heating capacity, by means of hot circulating liquids and forced air heat exchangers, to permit satisfactory operation of the vehicle and fire fighting systems for a 2-hour period at ambient temperatures as low as -40°F with the vehicle fully operational and the engine running. At the end of this 2-hour period, the vehicle shall be capable of successfully discharging its agents. All compartments not winterized shall be marked on the interior of the compartment as not winterized and not to be used for storage.

2.1.5 Foreign object prevention. All loose metal parts, such as pins and intake/discharge valve caps and plugs, shall be securely attached to the vehicle with wire ropes or chains. "Dog tag" style beaded chains are not acceptable. Removable panels, if provided, shall be attached with captive fasteners.

2.1.6 Mobility and performance.

2.1.6.1 Operating terrain. The vehicle shall operate on paved roads, graded gravel roads, cross country terrain, and a sand (desert) environment. Cross country terrain shall consist of open fields, broken ground, and uneven terrain.

2.1.6.2 Gradeability. The fully loaded vehicle shall be able to ascend any paved slope up to and including 50%.

2.1.6.3 Side slope stability. The fully loaded vehicle shall be stable on a 30° side slope when tested in accordance with 6.4.1 of NFPA 414. If an adjustable height suspension system is provided, the suspension system may be set to the height normally used on hard pavement.

2.1.6.4 Cornering stability. The fully loaded vehicle shall be stable when tested in accordance with 6.3.2 of NFPA 414.

2.1.6.5 Acceleration. The fully loaded vehicle shall accelerate from 0 to 50 miles per hour (mph) within 25 seconds on a level, paved road when tested in accordance with 6.4.3 of NFPA 414.

2.1.6.6 Top speed. The fully loaded vehicle shall attain a minimum top speed of 70 mph on a level, paved road when tested in accordance with 6.4.4 of NFPA 414.

2.1.6.7 Pump and roll on a 40% grade. The fully loaded vehicle shall be capable of pump and roll operations on a paved, dry, 40% grade in accordance with 4.4.1.3 of NFPA 414.

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2.2 Weights and dimensions.

2.2.1 Overall dimensions. Overall dimensions shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. Overall dimensions shall not exceed Table I:

TABLE I. Dimensions

Length (inches)	335
Width (inches) [air transport]]	104
Height (inches) [operational]	130
Height (inches) [air transport]	105

2.2.2 Overall weights. Overall axle and vehicle weights shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. De-rating the axles shall not be allowed. Weight distribution shall be in accordance with 4.3.1 of NFPA 414. Overall weights shall not exceed Table II:

TABLE II. Weight distribution

Per axle (pounds)	13,000
Vehicle (pounds)	26,000

2.2.3 Angles of approach and departure. The fully loaded vehicle shall have angles of approach and departure of not less than 30°.

2.2.4 Air transportability. The vehicle shall be transportable on C-130, C-17, and C-5 aircraft in accordance with MIL-STD-1791. The air transportable configuration shall include all equipment, dry chemical, and forcing agent tanks full, and the water and foam systems drained. In all air transport configurations, the vehicle shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The vehicle shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The vehicle shall drive on and off the aircraft, negotiating the required maximum ramp angles without shoring.

2.2.4.1 Shoring. The vehicle shall be air transportable without shoring.

2.2.4.2 Axle weight. Axle weight shall not exceed 13,000 pounds in air transport configuration.

2.2.4.3 Tire pressure. Tire pressure shall be capable of being reduced to 100 pounds per square inch (psi) for air transportability.

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2.2.4.4 Equipment removal and reconfiguration. Preparation for air transport shall take no more than 1 hour for two people to perform. All equipment removed shall be stored on the vehicle; caps and plugs shall permit moving and storage in transport configuration.

2.2.4.5 Tie downs. The vehicle shall be symmetrically restrained during air and ground transport. Tie down points shall be rated at a minimum of 25,000 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the vehicle. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209.

2.3 Chassis and vehicle components.

2.3.1 Engine. The vehicle shall have a turbocharged diesel engine compatible with the fuels listed in 2.3.3 containing over 15 parts per million (ppm) sulfur.

2.3.1.1 Engine block heater. An immersion style engine block heater shall be provided.

2.3.1.2 High idle switch. A high idle switch, which does not increase engine speed more than 25% above normal low idle speed and does not exceed the engine manufacturer's recommendation, may be provided to increase alternator, air compressor, or air conditioning compressor output to meet the maximum load requirements. A lighted rocker or toggle switch, accessible from the driver's seated position, shall activate the high idle control unit. The high idle switch shall operate only when the vehicle is out of gear and the parking brake is engaged. The high idle control unit shall automatically disengage if the transmission is placed in gear, the parking brake is released, or the fire pump is engaged. The high idle control unit shall not engage automatically; the high idle control unit shall only engage when the high idle switch is in the "ON" position.

2.3.2 Engine cooling system. The engine cooling system shall be in accordance with 4.4.2 of NFPA 414. Radiator hoses, heater hoses, and clamps, shall be in accordance with engine manufacturer recommendations. A coolant filter shall be provided. A label shall be installed near the engine coolant reservoir reading “Engine Coolant Fill.”

2.3.3 Fuel system. The fuel system shall be in accordance with 4.4.3 of NFPA 414.

2.3.3.1 Engine primary fuels. The vehicle engine shall be capable of operating on the following primary fuels:

a. JP-8, in accordance with MIL-DTL-83133, -40° F to 125° F ambient air temperature.

b. F-24, in accordance with NATO Standard AFLP-3747, -40° F to 125° F ambient air temperature. (Jet A in accordance with either ASTM D1655 or Defense Standard 91-91 with JP-8 additives in accordance with MIL-PRF-25017.)

c. Jet A1 in accordance with either ASTM D1655 or Defense Standard 91-91 with JP-8 additives in accordance with MIL-PRF-25017

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2.3.3.2 Engine alternate fuels. The vehicle engine shall be capable of operating on the following alternate fuels:

a. Diesel Gas Oil in accordance with JIS K 2204, based on nationally set seasonal temperature requirements.

b. Automotive Diesel in accordance with TUPRAS 403, based on nationally set seasonal temperature requirements.

c. Automotive Fuel, Diesel in accordance with BS EN 590, based on nationally set seasonal temperature requirements.

d. Diesel Fuel Oil, in accordance with GOST 305, based on nationally set seasonal temperature requirements.

e. JP-5, in accordance with MIL-DTL-5624, -40° F to 125° F ambient air temperature.

f. TS1 in accordance with GOST 10227-86 with JP-8 additives in accordance with MIL-PRF-25017, -40° F to 125° F ambient air temperature.

2.3.3.3 Engine emergency fuels. The vehicle engine shall be capable of operating on Jet A, Jet A1 in accordance with ASTM D1655, all ambient temperatures, as an emergency fuel.

2.3.3.4 Fuel priming pumps. The vehicle shall be equipped with an electric fuel pump in addition to the mechanical fuel pump. The electric pump shall be used as a priming pump capable of re-priming the engine's fuel system.

2.3.3.5 Fuel filters. Primary and secondary fuel filters and a heated fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve that is readily accessible by an operator or a mechanic, and shall be in accordance with SAE J1839. A combination fuel filter and fuel/water separator may be provided. Fuel filter elements shall be easily replaceable by a mechanic without loss of engine prime.

2.3.3.6 Fuel tank(s). The vehicle shall have one or two fuel tanks with a minimum usable capacity in accordance with 4.4.3 of NFPA 414. Each tank shall have a fill opening of three inches minimum, readily accessible to personnel standing on the ground and designed to prevent fuel splash while refueling. Each tank shall be located and mounted so as to provide maximum protection from damage, exhaust heat, and ground fires. If more than one tank is furnished, means shall be provided to assure equalized fuel level in each tank. An overturn fuel valve shall be provided for each tank to prevent spillage in the event of a rollover. Each fuel tank shall have a label prominently displayed on the tank that reads "Diesel Fuel ONLY."

2.3.4 Exhaust system. The exhaust system shall be in accordance with 4.4.4 of NFPA 414. The exhaust system including the muffler shall be constructed of stainless steel. All components of the exhaust system including clamps, weather stops, bolts, nuts, fasteners, and specially designed components not constructed of a corrosion resistant stainless steel alloy shall require the application of a corrosion resistant finish. Exhaust system outlet shall be directed upward or to the rear, away from personnel accessing equipment compartments and the engine air intake, and shall not be directed toward the ground. The design of the exhaust system outlet shall preclude

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the entry of rain.

2.3.5 Transmission. A fully automatic transmission with a hydraulic torque converter and at least six forward speeds shall be provided. The normal driving range selector position shall provide at least five gear ratios without movement of the selector. The transmission shall be in accordance with 4.6 of NFPA 414.

2.3.6 Driveline. The vehicle driveline shall be in accordance with 4.6 of NFPA 414. If the driveline is equipped with a differential locking control, a warning/caution label shall be placed in view of the driver indicating the proper differential locking/un-locking procedures. The operator's manual shall also include a similar warning/caution.

2.3.7 Axle capacity. Each axle shall have a rated capacity, as established by the axle manufacturer, in accordance with 4.6.7 of NFPA 414.

2.3.8 Suspension. The suspension system shall be in accordance with 4.7 of NFPA 414.

2.3.9 Tires and wheels. Tires and wheels shall be in accordance with 4.8 of NFPA 414. The vehicle shall be equipped with single tires and wheels at all wheel positions. Tubeless steel belted radial tires with non-directional on/off-road type tread mounted on disc wheel assemblies shall be provided. Tire and wheel assemblies shall be identical at all positions. Tires and wheels shall be certified by the manufacturer for not less than 25 miles of continuous operation at 60 mph at the normal operational inflation pressure. A spare tire and wheel assembly shall be provided; however, the spare tire and wheel assembly is not required to be mounted on the vehicle. Tires shall be new. Retreads, recaps, or re-grooved tires are not acceptable. Tire valve stem caps shall be made of plastic. The vehicle tire pressure shall be indicated on a label installed above each wheel well.

2.3.10 Towing connections. The vehicle shall be equipped with towing connections in accordance with 4.9 of NFPA 414. The towing connections shall be designed to allow flat towing of the vehicle; the capability to lift and tow the vehicle is not required.

2.3.11 Brake system. The brakes shall be fully air-actuated, outboard mounted disc type. Brakes shall be in accordance with Code of Federal Regulations (CFR) 49 CFR 393.40 through 393.42(b)), 393.43, and 393.43 through 393.52. The vehicle shall be equipped with a multi-channel all-wheel antilock brake system. The braking system shall be in accordance with 4.10 of NFPA 414 and be provided with all necessary components to include:

a. Air compressor having a capacity of not less than 16 standard cubic feet per minute (scfm).

b. Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between the compressor and the reservoir tank.

c. Automatic moisture ejector on each air storage reservoir. Manual air tank drains are an acceptable alternative if they are labeled and are located in one compartment, accessible by an individual standing at the side of the vehicle.

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d. Brakes shall be self adjusting.

e. Spring set parking brakes.

f. All components of the braking system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of wheel rims, to ensure, in case of a flat tire, that the weight of the vehicle will be supported by the rim and the flat tire and not be imposed on any component of the braking system. Slack adjusters and air chambers shall be located above the bottom edge of the axle carrier.

2.3.11.1 Air dryer. A replaceable cartridge desiccant air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95% of the moisture in the air being dried. The dryer shall have a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

2.3.11.2 Compressed air shoreline. A checked, auto-eject compressed air shoreline connection shall be provided to maintain brake system pressure while the vehicle is not running. The shoreline shall be flush mounted (not to extend outside the body line), located on the exterior of the vehicle at the rear. A minimum 50 foot long air supply hose equipped with an appropriate mating shoreline connector and an air fitting shall be provided with the vehicle.

2.3.11.3 Auxiliary air compressor. An auxiliary air compressor shall be provided to maintain vehicle air system working pressure to maintain air brake tanks.

2.3.11.4 Compressed air electrical shoreline connection. The auxiliary air compressor shall be powered from a covered, polarized, insulated, labeled, recessed, male, 110 volt AC auto-eject connection. The connection shall be located on the exterior of the vehicle at the rear. A weatherproof charge meter shall be installed next to the connection. A 50 foot long, three wire, 15 amp rated, 110 volt AC power cable, with straight blade (non twist-lock) connectors, shall be provided. This connection shall also be used to power the battery charger/conditioner in 2.9.3. When specified (see 6.2), the connection and power cable shall be for 220 volts in lieu of 110 volts.

2.3.11.5 Air hose reel. An air hose reel shall be provided in a compartment on the side of the vehicle equipped with 200 feet of 3/8-inch internal diameter (I.D.) hose. A 3/8-inch National Pipe Taper (NPT) fitting and female style quick disconnect shall be connected to the end of the hose. A 4-way roller guide shall be provided on the hose reel to prevent hose chafing and kinking. The hose line shall be equipped with a rubber ball stop to prevent hose pull through on roller guides during rewinding operations. A pressure protected air supply from the chassis air system shall be connected to the hose reel. The air supply lines shall be routed with minimum bends and located or guarded from damage from the carried equipment. The hose reel shall have electric rewind provisions with all electrical components sealed against entry of water. Manual rewind provisions shall also be provided with a manual rewind handle securely mounted in a compartment near the hose reel.

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2.3.12 Steering. The vehicle shall be equipped with power steering.

2.3.12.1 Steering effort. The steering system performance shall be in accordance with 4.11 of NFPA 414.

2.3.12.2 Vehicle clearance circle. The fully loaded vehicle shall have a vehicle clearance circle diameter in both directions which is less than three times the maximum overall length of the vehicle.

2.3.13 License plate bracket. A lighted license plate bracket shall be provided at the left rear of the vehicle.

2.4 Cab. The vehicle shall have a fully enclosed two door cab of corrosion resistant construction. Cab door openings shall extend for the full vertical height of the side panels. Steps and handrails shall be provided for all crew doors, and at least one grab handle shall be provided for each crew member, located inside the cab for use while the vehicle is in motion. The lowermost step(s) shall be no more than 22 inches above level ground when the vehicle is fully loaded. A tilt and telescoping steering column shall be provided.

2.4.1 Windshield and windows. The windshield and windows shall be of tinted safety glass. A watertight hatch shall be provided in the cab roof to facilitate emergency occupant escape in the event of a vehicle accident. A label reading “Emergency Escape Hatch” shall be installed on the cab interior side of the hatch. The hatch shall be painted to reduce reflection and outside temperatures from heating the cab area in high operating temperature environments.

2.4.2 Cab interior sound level. The maximum cab interior sound level shall be in accordance with 4.12.3.3 of NFPA 414.

2.4.3 Instruments and controls. All instruments and controls shall be illuminated. Gauges shall be provided for engine oil pressure, engine coolant temperature, and automatic transmission oil temperature. All switches and/or controls that activate systems or system components shall be labeled as to their function. In addition to the instruments and controls required by 4.12.4.4 and 4.12.4.5 of NFPA 414, the following shall be provided within convenient reach of the seated driver:

a. Master warning light control switch,

b. Work light switches for side and rear light emitting diode (LED) work lights,

c. Compartment "Door Open" warning light and intermittent alarm that sounds when a compartment door is open and the transmission is in any gear other than neutral, and

d. An engine hour meter.

e. Pump hour meter.

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2.4.4 Windshield deluge system. The vehicle shall be equipped with a powered windshield deluge system in accordance with 4.12.4.6 of NFPA 414. The deluge system shall be supplied from the agent water tank and shall have an independent pumping system. The deluge system activation switch shall be located within reach of the seated driver and the turret operator. The deluge system shall include a replaceable filter and a manual valve for use when the filter is changed.

2.4.5 Forward looking infrared. The vehicle shall be equipped with a forward looking infrared (FLIR) system with camera and monitor in accordance with 4.12.4.8 of NFPA 414. The FLIR monitor described in 4.12.4.4 (14) of NFPA 414 shall have a minimum dimension of 10-inches (measured diagonally). The monitor shall be so located as to be visible to both the seated driver and turret operator.

2.4.6 Climate control system. The contractor's standard heater/defroster and air conditioning system shall be provided. In 125°F ambient temperature and at maximum compressor speed, the air conditioning system shall cool the fully occupied cab to 75°F within 45 minutes. The climate control system shall induct at least 60 cubic feet per minute of fresh air into the cab. At least four adjustable panel louvers shall be provided, approximately evenly spaced across the width of the cab; each shall provide approximately equal flow. Cab mounted components shall be protected from inadvertent damage by personnel.

2.4.7 Seats. The driver seat and the turret operator's seat to the driver's right shall each be, capable of being adjusted fore and aft and for height. The driver's seat shall also be equipped with a recline feature. A third fixed (non-suspension) seat for an additional crew member shall be located to the left, and possibly behind the driver. The turret operator and the crew member's seats shall both be provided with a backrest, parade cover, and a bracket designed to store a one hour capacity self-contained breathing apparatus (SCBA).

2.4.8 Seat belts. Each seat shall be provided with a Type 2 seat belt assembly (that is, 3-point restraint) in accordance with 49 CFR 571.209. A seat belt which is an integral part of the seat with no remote mounting point shall be provided. Seat belts shall be of sufficient length to accommodate personnel in full personal protective equipment (PPE). Seat belts shall be red in color.

2.4.9 Seat belt warning system. A seat belt warning system in accordance with 14.1.3.9 of NFPA 1901 shall be provided.

2.4.10 Windshield wipers and washer. The vehicle shall be equipped with electrically powered windshield wipers. The wiper arms and blades shall be of sufficient length to clear the windshield area described by SAE J198. Individual wiper controls shall include a minimum of two speed settings and an intermittent setting. The wiper blades shall automatically return to a park position, out of the line of vision. The vehicle shall be equipped with a powered windshield washer system, including an electric fluid pump, a minimum 1-gallon fluid container, washer nozzles mounted to the wiper arms (wet arms), and a momentary switch.

2.4.11 Warning signs. Signs that read "Occupants must be seated and belted when apparatus is in motion" shall be provided in locations that are visible from each seated position in accordance with 4.12.6 of NFPA 414.

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2.4.12 Lateral acceleration indicator. The vehicle shall be equipped with a lateral accelerometer in accordance with 4.12.8 of NFPA 414.

2.4.13 Field of vision. The vehicle shall have a field of vision in accordance with 4.3.2.2 of NFPA 414.

2.4.14 Mirrors. Combination flat and convex outside rearview mirrors shall be installed on each side of the cab. The flat mirrors shall be of the motorized remote control type, providing not less than 60° horizontal rotational viewing range. The flat mirrors shall also have electrically heated heads. Mirror remote and heating controls shall be located on the instrument panel within reach of the seated driver. To provide the driver a clear view of the area ahead of the vehicle, a rectangular mirror shall be installed on the lower corner of each side of the windshield, but not within the driver's direct line of sight, having a minimum area of 35 square inches.

2.4.15 Backup camera. A rear vision camera and in-cab monitor shall be provided to aid the driver in safely backing up the vehicle. The monitor shall be located to the right side of the driver. A switch shall be provided to allow the driver to manually activate the back-up camera from within the cab. The back-up camera shall also be switched "on" automatically whenever the vehicle is in the reverse mode of operation. The backup camera and FLIR camera shall use the same monitor.

2.5 Body, compartments, and equipment mounting.

2.5.1 Body. The vehicle shall have a corrosion resistant body.

2.5.2 Compartments. The vehicle body shall have lighted compartments in accordance with 4.13.3 of NFPA 414. A minimum of 50 cubic feet of enclosed storage space shall be provided, sufficient to transport/store fire fighting and PPE listed in 2.5.5.

2.5.2.1 Compartment doors. All compartments shall have counterbalanced, non-locking, roll-up doors and shall be painted Desert Sand. Door latch handles shall be full-width bar type. Door straps shall be provided to assist in closing the compartment doors when the door height exceeds six feet above the ground.

2.5.2.2 Scuffplates. Replaceable scuffplates shall be provided at each compartment threshold to prevent body damage from sliding equipment in and out of the compartments. The scuffplates shall be securely attached to the compartment threshold but shall be easily replaceable in the event of damage.

2.5.2.3 Drip rails. Drip rails shall be provided over or incorporated in each compartment door. If the drip rails are not integral with the body, they shall be of anodized extruded aluminum and shall be painted Desert Sand.

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2.5.2.4 Trays. A height adjustable roll-out tray shall be provided for every 18-inches of each vertical storage compartment door opening with each tray capable of being removed. Height adjustments shall require no more than common hand tools, and shall not require disassembly of fasteners. Each tray shall support a minimum of 500 pounds without experiencing permanent deformation. Each tray shall be accessible to crew members standing on the ground. A pull out and tip-down configuration shall be used for trays over 54-inches from the ground when the compartment is more than 18-inches deep. Each tray shall have a ¼-inch diameter drain hole located in each corner to allow drainage of water.

2.5.2.5 Drainage mats. Each compartment floor and shelf shall be covered with a removable black mat designed to allow for drainage of any water from the stowed equipment.

2.5.3 SCBA storage tubes. Tubes for storing four spare SCBA 1-hour bottles shall be provided. The tubes shall be in accordance with 15.5 of NFPA 1901 requirements and shall be of sufficient size to accommodate any commercially available 1-hour SCBA cylinder and valve head.

2.5.4 Ladder, handrails, and walkways. Ladders, stepping, standing, and walking surfaces shall be in accordance with 4.13.6 of NFPA 414. Handrails shall be provided in accordance with 4.13.5 of NFPA 414. The lowermost step(s) or ladder rungs shall be no more than 22 inches above level ground when the vehicle is fully loaded. The vehicle shall have a rear ladder for personnel to gain access to the roof of the vehicle.

2.5.5 Ancillary equipment. The following equipment items shall be furnished. These items shall be inventoried, packaged, and shipped as loose equipment. All necessary brackets and hardware shall be provided to allow the items to be mounted by the receiving installation after delivery.

a. Two dual spanner wrench holders.

b. Four spanner wrenches.

c. Two hydrant wrench holders.

d. Two hydrant wrenches.

e. One rubber mallet, 24 ounce, 15-inch overall length, 2-inch diameter, 4-inch long head.

f. Two couplings, hose, double male, swivel, internal 2½-inch National Hose Thread (NHT) hard coated aluminum with rocker lug.

g. Two adapters, double female, swivel, 2½-inch NHT, aluminum.

h. One gated wye connection, 2½-inch NHT female to two 1½-inch NHT male.

i. One 20 foot length of 6-inch soft suction hose with 4½-inch NHT female thread connections.

j. One axe with a fiberglass handle, flathead and blade shield and mounting bracket.

k. One axe with a fiberglass handle, pickhead and blade shield and mounting bracket.

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l. One extinguisher, 2½-gallon pressurized water and mounting bracket.

m. One extinguisher, 20 pound, dry chemical and mounting bracket.

n. One set of folding wheel chocks and mounting brackets.

o. One set of Storz wrenches and holders, 4-inch to 4½-inch.

p. One emergency road kit consisting of at least the following items:

1. Two highway warning triangles.
2. Two reflective vests.
3. Two cyalume light sticks.
4. One water resistant storage bag.

q. Two 10 foot long, light weight 5-inch hard suction hoses, with 4½-inch NHT long handle female threads on one end and 4½-inch NHT long handle male threads on the other. The hoses shall be secured in brackets mounted on top of the vehicle or stored in a recessed compartment on drainable trays.

r. One chrome plated barrel suction strainer with 4½-inch NHT male thread connections.

2.6 Agent system.

2.6.1 Agent pump. The vehicle shall be equipped with a pump capable of providing the performance specified herein. The pump shall be capable of delivering 1250 gallons per minute (GPM) of foam solution at 225 PSI at the bumper turret nozzle and 95 GPM of foam solution at 100 PSI to the handline nozzle. The pump shall have the capability of providing agent to the bumper turret and handline simultaneously. The system shall include provisions to prevent foam from being relieved onto the ground. Provisions shall be made to prevent pump cavitation. For examples the installation of a low pressure switch in the pump inlet to shut down operation in the event of loss of suction pressure. The pumping system shall be capable of providing discharge at full flow within 1.5 seconds of valve actuation. The pump and gear box shall have a cooling system that allows the pump to be continuously operated if attached to an alternate water source

2.6.1.1 Agent system piping. All metallic surfaces of the piping and associated components that come into contact with the agent shall be of brass, bronze, or passivated stainless steel.

2.6.1.2 Flushing system. A flushing system shall be provided to purge all piping and associated components with clear water.

2.6.1.3 Tank to pump connection. A check valve and shutoff valve shall be provided in each tank to pump line.

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2.6.1.4 Intake connections. The vehicle shall be equipped with one valved male 4½-inch NHT intake connection on the left side. The vehicle shall be equipped with one valved female 2½-inch NHT intake connection on the left side fitted with a 30° turn-down fitting, adjacent to the 4½-inch intake connection. The 4½-inch intake connection shall have a quarter-turn control valve, a bleeder valve, a strainer, a cap, and a slow-operating valve and an automatic pressure relief device in accordance with 16.6 of NFPA 1901. The 2½-inch intake connection shall have a quarter-turn control valve, a bleeder valve, a strainer, and a plug in accordance with 16.6 of NFPA 1901. The vehicle shall be capable of filling its water tank by pumping from a draft, a hydrant, or a nurse truck through any of the intake connections without having to connect a hose from a discharge connection to a tank fill connection.

2.6.1.5 Discharge connections. One 2½-inch male NHT discharge connection shall be provided on each side. Each connection shall be equipped with a cap, a quarter-turn control valve, a bleeder valve, a pressure gauge and a 30° turn-down fitting. Each connection shall be rated at 250 GPM minimum.

2.6.1.6 Piping, couplings, and valves. The agent system piping shall incorporate groove type couplings to the maximum practical extent. All piping between agent system components shall be readily removable without disturbing the components.

2.6.1.8 Overheat protection. The agent system shall be equipped with an overheat protection system in accordance with 4.15.6 of NFPA 414.

2.6.1.9 Pressure relief valves. The agent system shall be equipped with pressure relief valves in accordance with 4.15.7 of NFPA 414.

2.6.1.10 Drains. The agent system shall be equipped with a drainage system in accordance with 4.15.8 of NFPA 414.

2.6.1.11 Priming device. An electro/mechanical priming pump or a venturi device shall be provided to allow pump operation from draft for discharge or water fill operations.

2.6.2 Water tank. The vehicle shall have a baffled water tank with a manufacturer certified minimum usable capacity of at least 1,000 gallons.

2.6.2.1 Water tank construction. The water tank shall be constructed of UV resistant polypropylene and have a lifetime warranty. All materials used shall be capable of storing water, foam concentrate, and water/foam solutions.

2.6.2.2 Water tank manhole cover and drain. The water tank shall be equipped with an easily removable manhole cover having a minimum opening diameter of 20-inches. The water tank shall incorporate a drain line with a drain valve capable of being controlled by an individual standing next to the vehicle. The drain line shall have a minimum 2-inch I.D. with its discharge point to be below the under body panels.

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2.6.2.3 Water tank overflow system and venting. The water tank shall incorporate a vent system to relieve pressure on the tank during fill and discharge operations at maximum flow rates and an overflow system to relieve excess liquid in the event of tank overfill. The vent and overflow system shall prevent leakage of water when the tank is filled to capacity and the vehicle is operating on the maximum side slopes and grades specified herein. Drainage from the vent and overflow system shall not flow over body panels or other vehicle components and shall not be in the track of any of the tires. Tank vent hoses shall be of the non-collapsible type.

2.6.2.4 Water tank top fill opening. A top fill opening of not less than 8-inches I.D. with a readily removable ¼-inch mesh strainer shall be provided. The fill opening may be incorporated as part of the manhole cover, and shall be sized to accommodate a 2½-inch fill hose. The opening shall be marked "Water Fill."

2.6.2.5 Water tank fill connections. The water tank shall incorporate a 4½-inch male NHT connection and a 2½-inch female NHT rocker lug connection on each side of the vehicle. Each connection shall be fitted with a 30° turn-down fitting. The water fill connections shall allow external re-supply of the water tank during discharge pumping operations with each connection to be in accordance with 4.16.3 of NFPA 414.

2.6.3 Foam system.

2.6.3.1 Foam concentrate tank. The foam concentrate tank shall have a manufacturer certified working capacity sufficient for two tanks of water at the maximum tolerance specified in NFPA 412, *Standard for Evaluating Aircraft Rescue and Fire-Fighting Foam Equipment* for 6-percent foam concentrate (that is, 7.0-percent).

2.6.3.1.1 Foam tank construction. The foam tank shall be constructed of UV resistant polypropylene and have a lifetime warranty. All materials used in construction shall be capable of storing foam concentrate.

2.6.3.1.2 Foam tank drain. The foam tank shall incorporate a drain line with a drain valve capable of being controlled by an individual standing next to the vehicle. The drain line shall have a minimum 1½-inch I.D. and be located to allow the contents of the tank to be drained into 5-gallon cans or 55-gallon drums.

2.6.3.1.3 Foam tank top fill trough. The foam tank shall incorporate a top fill trough mounted in the top of the tank readily accessible on top of the vehicle. The top fill trough shall incorporate a cover and latch, sealed so as to prevent spillage under any operating condition. The top fill trough neck shall extend sufficiently close to the bottom of the tank to reduce foaming to a minimum during the fill operation. The top fill trough shall incorporate readily removable, rigidly constructed 10 mesh stainless steel or brass strainers. All components in and around the top fill trough shall be constructed of materials that resist all forms of deterioration that could be caused by the foam concentrate or water. The opening shall be marked "Foam Fill."

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2.6.3.2 Foam tank fill connections. The foam tank shall incorporate a 1½-inch female NHT connection on each side of the vehicle to permit filling by an external transfer hose at flow rates up to 7.5 GPM. The connections shall be provided with chained-on long handled plugs or rocker lug plugs. The top of the connections shall be no higher than 48-inches above the ground and readily accessible. The fill lines shall incorporate check valves and readily removable, rigidly constructed ¼-inch mesh strainers. All components in the foam tank fill system shall be constructed of materials that resist all forms of deterioration that could be caused by the foam concentrate or water.

2.6.3.2.1 Foam tank vent and overflow system. The foam tank shall incorporate a vent system to relieve pressure on the tank during fill and discharge operations at maximum flow rates and an overflow system to relieve excess liquid in the event of tank overfill. The vent and overflow system shall prevent leakage of foam when the tank is filled to capacity and the vehicle is operating on the maximum side slopes and grades specified herein. Drainage from the vent and overflow system shall not flow over body panels or other vehicle components and shall not be in front of or behind any of the tires. Tank vent hoses shall be of the non-collapsible type.

2.6.3.3 Foam transfer pump. An electric motor driven or pneumatic, self-priming, pump shall be provided and mounted in a compartment on the vehicle. The pump shall be driven from the vehicle electrical or air system. The pump shall be capable of transferring and drawing foam liquid concentrate into the bottom of the foam tank at up to 25 GPM directly through a 1½-inch NHT female hose thread connection. All materials and components that come in contact with the foam shall be compatible with the foam concentrate. The pump and its plumbing shall have provisions for flushing with water from the water tank. The pump shall be removable as an assembly without disturbing other components. A 6 foot length of hose with a 1½-inch NHT male hose thread connection on one end shall be provided for attachment to the pump connection when filling the foam tank from an external foam storage container.

2.6.3.4 Foam flushing system. The foam concentrate system shall be designed in accordance with 4.17.2 of NFPA 414 so that the system can be readily flushed with clear water.

2.6.3.5 Foam liquid concentrate piping. All metallic surfaces of the piping and associated components that come into contact with the foam concentrate shall be of brass, bronze, or passivated stainless steel. The foam concentrate piping shall be in accordance with 4.17.3 of NFPA 414.

2.6.4 Foam proportioning system. The vehicle shall have an around-the-pump or a direct injection foam proportioning system for Aqueous Film-Forming Foam (AFFF). The system shall automatically and uniformly proportion water and 3-percent foam concentrate within a minimum ratio of 2.8-percent to a maximum ratio of 3.5-percent foam concentrate to water by volume. The system shall also be capable of proportioning 6-percent foam concentrate (5.6-percent minimum, 7.0-percent maximum). If a fixed orifice plate system is used, a plate shall be provided for each percentage foam concentrate; the additional plate shall be securely mounted in a protected location on the vehicle. A mechanic shall be able to interchange the plates using common hand tools.

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2.6.4.1 Foam testing system. The vehicle shall have a foam testing system featuring water only plumbing to accurately test foam percentages without the use of foam. The system shall be designed to simulate AFFF concentrate for test purposes. Actual solution flow shall be measured on each vehicle based on the flow accuracy requirements in NFPA 414 (rated flow -0/+10%). The system shall check for foam percentage accuracy on each discharge in accordance with NFPA 412. The acceptable range for 3% foam shall be 2.8-3.5% for turrets and ground sweeps and 2.8-4.0% for handlines and under truck, and for 6% foam shall be 5.5-7.0% for turrets and ground sweeps and 5.5-8.0% for handlines and under truck. The vehicle shall have an indicator warning when the truck is in test mode. The foam testing system shall be capable of being operated from the cab or structural panel. The test system shall be disengaged by means of a power off cycle. The system shall be capable of operation with standard pressure system. The system shall have a visual reading to show the percentage foam. The actual results shall be downloadable for records retention. The system shall operate in all temperatures across the operating temperature range. All piping shall be made of passivated stainless steel, grade 304/304L or 316/316L.

2.6.5 Primary turret. The vehicle shall be equipped with a primary turret mounted on the front bumper in accordance with 4.19.1 of NFPA 414.

2.6.5.1 Bumper turret. The vehicle shall be equipped with a joystick controlled, constant flow, non-air-aspirating, variable stream type bumper turret. The bumper turret shall be capable of discharging with dual flow rates of foam or water rated at 375/750-gpm minimum, with a pattern infinitely variable from straight stream to fully dispersed. The nozzle pressure for the bumper turret shall be in accordance with 2.6.1. The bumper turret shall be capable of automatic oscillation, with the range of oscillation fixed at 45° each side of center (left and right) and elevation/depression.

2.6.6 Handline.

2.6.6.1 Handline. A handline equipped with 150 feet of 1-inch hose shall be provided in a lower compartment on the side of the vehicle. The handline shall be capable of discharging 95 GPM of water/foam solution at a minimum nozzle pressure of 100 PSI. The minimum straight stream discharge range shall be 65 feet with the minimum foam quality expansion ratio specified and tested in accordance with NFPA 412.

2.6.6.2 Handline nozzle. The handline nozzle shall be a variable stream pistol grip type, capable of providing a discharge pattern from straight stream to fully dispersed (fog).

2.6.6.3 Handline hose reel and storage. The handline hose storage area shall be fabricated from non-corrosive material and shall be designed to drain effectively. The handline storage area shall be smooth and free from all projections that might damage the hose. No equipment shall be mounted or located where it could obstruct removing the hose. An agent discharge switch and a blow down switch to purge the hose of agent shall be mounted on the hose reel. The hose reel shall have electric rewind provisions with all electrical components sealed against entry of water. Manual rewind provisions shall also be provided with a manual rewind handle securely mounted in a compartment near the hand line. A control for charging the handline shall be provided on the instrument panel accessible to the driver and the turret operator.

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2.7 Structural panel. A limited structural fire fighting system shall be provided. The system shall include a control panel mounted in a compartment on the left side of the vehicle capable of activation by an individual standing next to the panel. An interlock shall be provided which will allow structural system operation only when the vehicle parking brake is set and the transmission is in the neutral position. Controls and instruments on the panel shall be grouped by function with the panel to be either hinged or accessible from the rear for maintenance. Instruments shall be lighted for night operation. The structural panel shall include, as a minimum, the following:

a. Switch to activate panel and panel lights.

b. Water mode selection switch.

c. Foam mode selection switch.

d. Manual engine throttle with micro adjustments and pilot controlled relief valve or an electronic pressure sensing governor.

e. OK to pump from panel indicator light.

f. Engine tachometer.

g. Engine oil pressure gauge.

h. Engine coolant temperature gauge.

i. Liquid filled pump suction pressure gauge, -30 inches Hg vacuum to 400 psi.

j. Liquid filled pump discharge pressure gauge, 0 to 400 psi.

k. Fill water tank from draft switch.

l. Water tank drain switch.

m. Water tank shutoff switch.

n. Priming device control.

2.8 Dry chemical agent system. A pressure indicator shall be visible to any person opening the tank fill cap. Blow-down piping shall be directed beneath the vehicle. The dry chemical agent tank shall include lifting rings and shall have a nameplate indicating, as a minimum, the following:

a. Extinguishing agent.

b. Capacity.

c. Weight full.

d. Weight empty.

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e. Operating pressure.

f. Hydrostatic test date.

g. Type of agent required for re-servicing.

2.8.1 Dry chemical container. A dry chemical container having a capacity of 250 pounds of potassium bicarbonate dry chemical auxiliary agent shall be provided. The container shall be in accordance with 4.22.1 of NFPA 414.

2.8.2 Dry chemical propellant cylinder. The propellant gas cylinder for the dry chemical system shall be in accordance with 4.22.2 of NFPA 414. The cylinder shall be secured to withstand off-road operations and be replaceable within fifteen minutes by two crew members standing next to the vehicle.

2.8.3 Dry chemical hose reel. A hose reel equipped with 150 feet of 1-inch dry chemical hose and a dry chemical nozzle shall be mounted in a compartment. The dry chemical nozzle shall be capable of a discharge rate of 5 lb/sec. A dry chemical discharge switch and a blow down switch to purge the hose of dry chemical shall be mounted on the hose reel. The hose reel shall have electric rewind provisions with all electrical components sealed against the entry of water. Manual rewind provisions shall also be provided with a manual rewind handle securely mounted in a compartment near the hand line. A quick acting control to activate the hand line from the cab shall be provided.

2.9 Electrical system and warning devices. The vehicle shall have an electrical and starting system in accordance with 4.5 of NFPA 414. All electrical components within the cab shall comply with NEMA 2 or IP51s standards, and all exterior electrical components shall comply with NEMA 4 or IP65 standards.

2.9.1 Alternator. A single or dual alternator charging system in accordance with 13.3 of NFPA 1901 shall be provided. The minimum continuous electrical load shall include operation of the air conditioning system.

2.9.2 Batteries. Batteries shall be of the commercial maintenance-free sealed lead acid, starved electrolyte, gas recombination, spiral wrapped, absorbent gas mat (AGM), top post type in accordance with MIL-B-18013/1.

2.9.2.1 Battery compartment. The batteries shall be enclosed in a weatherproof box or compartment and be readily accessible.

2.9.3 Battery charger or conditioner. The vehicle shall have a DC taper type battery charger or an automatic battery conditioner, providing a minimum 12 amp output. The charger/conditioner shall be permanently mounted on the vehicle in a ventilated, accessible location. The charger/conditioner shall be powered from the electrical shoreline connection (see 2.9.3.1). A charging indicator shall be installed next to the connection. When a battery conditioner is provided, the conditioner shall monitor the battery state of charge and, as necessary, automatically charge or maintain the batteries without gassing, depleting fluid level, overheating, or overcharging.

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2.9.3.1 Electrical shoreline connection. When specified (see 6.2), the connection and power cable shall be for 220 volts in place of 110 volts. The battery charger/conditioner shall be powered from a covered, three wire, straight blade, polarized, insulated, labeled, recessed, 120 volt, NEMA 5-30P male plug inlet, in a weatherproof AC auto-eject receptacle. It shall be located on the exterior of the vehicle, either adjacent to the driver's door or within 6 -inches of the left side front corner of the body. The shoreline connection's location shall not pose a tripping hazard to pedestrians walking by the vehicle when the shore line is connected. A weatherproof charge meter shall be installed next to the receptacle. A 50 foot long, three wire, 30 amp rated, 120 volt, AC power cable, with straight blade (non twist-lock), NEMA 5-30R and 5-30P style, in accordance with NEMA WD-6, connectors, shall be provided. The winterization system shall have a labeled activation switch with a pilot light.

2.9.3.2 Slave start receptacle. A slave start receptacle shall be provided in a location accessible to an individual standing next to the vehicle.

2.9.3.3 Jump start connections. Jump start connections shall be provided on the exterior of the battery box.

2.9.4 Electromagnetic interference. The vehicle shall be in accordance with the following conducted and radiated emission and susceptibility requirements of MIL-STD-461: CE102, CE106, CS101, CS103, CS104, CS105, CS114, CS115, CS116, RE102, and RS103.

2.9.5 Work lighting.

2.9.5.1 Cab interior lights. Cab interior light levels shall be sufficient for reading maps or manuals. Two sets of red and white LED cab interior dome lights shall be provided, with one set mounted above the driver and the second set mounted above the turret operator.

2.9.5.2 Compartment lights. LED strip lighting shall be placed from top to bottom along each side of a compartment equipped with a roller shutter door. Any compartment not having a roller shutter door shall be provided with LED lighting. Compartment lighting shall automatically illuminate only when a compartment door is opened and the vehicle electrical master switch is in the "on" position. Light switches shall be of the magnetic (non-mechanical) type.

2.9.5.3 Ladder, step, walkway, and area lights. Non-glare white LED lighting shall be provided at ladders and access steps where personnel work or climb during night operations. LED ground lighting in accordance with 13.10.1 of NFPA 1901 shall be provided. These area lights shall be controlled with three-way switches on the cab instrument panel and near the light sources. The switch located in the cab shall be a master switch and shall be turned on before auxiliary switches near the light sources are operational. The ground lights shall be controlled by a master switch located on the cab instrument panel as well as by switches on the cab doors.

2.9.5.4 Scene lights. A total of six LED scene lights shall be provided to illuminate the work areas in front of and on each side of the vehicle. Two scene lights shall be mounted above the windshield with two scene lights mounted on each side on the front and rear corners of the center body area. A switch shall also be mounted on the instrument panel to control the front scene lights. Each pair of side mounted lights shall be controlled by a switch on the instrument panel as well as by a switch on that side. All trucks shall be equipped with a steady red front warning light as required by California state law regardless of shipping destination.

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2.9.6 Audible warning devices.

2.9.6.1 Siren. The vehicle shall be equipped with an electronic siren system. The amplifier unit shall include volume control and selection of "Radio," "PA," "Manual," "Yelp," "Wail," and "Hi-Lo" (European) modes, and a magnetic noise canceling microphone. The amplifier, microphone, and controls shall be within reach of the driver and the turret operator. Siren activating foot switches shall be located in front of the driver and the turret operator. The siren speaker shall be rated at 100 watts minimum and shall be located in a guarded position as low and as far forward on the vehicle as practical.

2.9.6.2 Horns. Dual forward facing air horns shall be installed in protected locations near the front of the vehicle. Air horn activating foot switches shall be located in front of the driver and the turret operator.

2.9.7 Emergency warning lights. All emergency warning lights shall use LED elements. Two identical minimum 22-inch wide LED emergency warning light bars shall be mounted on top of the vehicle, directly on or immediately behind the cab, with one angled toward the outside on each side. Each light bar shall have a minimum of twelve LED elements with one element at each corner, three at the front, three at the rear and one on each side. Two round LED emergency warning lights having a minimum diameter of 6-inches and a minimum height of 5-inches shall be provided, one installed on each top rear corner of the vehicle. Three LED emergency warning lights shall be provided on each side, one mounted as far forward, one mounted as far to the rear, and one mounted as close as practical to the mid-point of the vehicle. Two LED emergency warning lights shall be mounted on the front of the vehicle. Two LED emergency warning lights shall be mounted on the rear of the vehicle. A single switch shall be provided on the instrument panel to control all of the top, side, front and rear emergency warning lights. A switch shall also be provided on the instrument panel to disable all ten lower emergency warning lights when desired.

2.9.7.1 Emergency warning light color. All emergency warning lights shall be red. When specified (see 6.2), the rearward, red emergency warning lights shall be replaced with amber. When specified (see 6.2), all red emergency warning lights shall be replaced with blue.

2.9.7.2 Headlight flashing system. A high beam, alternating/flashing, headlight system shall be provided. The headlight flasher shall be separately switched from the warning light panel.

2.9.8 Radio circuits. The vehicle shall have two separate 30 amp circuits, with breakers and at least 6 feet of wire routed to a space provided adjacent to the driver and turret operator for purchaser provided radios and other electrical equipment. The wiring shall be tagged indicating its purpose. Two antennas shall be installed on the roof of the cab, pre-wired to the space in the cab where purchaser provided radios would be installed.

2.9.9 Auxiliary power receptacles. The vehicle shall have two 12 volt AC auxiliary power receptacles mounted adjacent to the driver and crew member positions, preferably in the instrument panel.

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2.10 Workmanship. The vehicle, including all parts and accessories, shall be fabricated in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, burrs, defects, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of welding, brazing, soldering, riveting, and painting; alignment of parts; tightness of fasteners. The vehicle shall be thoroughly cleaned of all foreign matter.

2.11 Training. A minimum of two days of training shall be provided at the receiving base following delivery of the vehicle. Training shall consist of driver/operator training, and informational overview training for maintenance personnel.

2.12 Delivery. All trucks shall be delivered via flat bed or low boy.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production vehicle inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First production vehicle inspection. The first production vehicle shall be subjected to the examinations and tests described in 4.6.3.1.1 through 4.6.3.1.4, 4.6.3.1.6, 4.6.3.1.7, 4.6.3.1.10, 4.6.3.1.13, 4.6.3.2, 4.6.3.3.4 through 4.6.3.3.6, 4.6.3.3.8, 4.6.3.4.1 and 4.6.3.4.2. The contractor shall provide or arrange for all test equipment, personnel, and facilities.

4.3 Conformance inspection. Each production vehicle shall be subjected to the examinations and tests described in 4.6.3, 4.6.3.1.5, 4.6.3.1.8, 4.6.3.1.9, 4.6.3.1.11, 4.6.3.1.12, 4.6.3.3.1 through 4.6.3.3.3, 4.6.3.3.7 and 4.6.3.5.

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4.4 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

4.5 Commercial item requirement. The vehicle furnished shall comply with the "commercial item" definition of FAR 2.101 as of the date of award. The Government reserves the right to require the manufacturer to prove that their product complies with the referenced commerciality requirements and each salient characteristic of this CID. The manufacturer shall provide an itemized technical proposal that describes how the proposed model complies with each salient characteristic of this CID; a paragraph by paragraph response to the salient characteristics paragraph of this CID shall be provided. The response to each paragraph shall clearly state “Fully Comply” or “Bid with Exception” in addition to other data the contractor wishes to submit. The proposal shall be provided with the pricing submission. Failure to provide this information may deem a vendor as non-responsive and their proposal may be rejected. The manufacturer shall provide two copies of their commercial descriptive catalogs with their offer as supporting reference to the itemized technical proposal. The manufacturer shall identify all modifications made to their commercial model in order to comply with the requirements herein.

4.6 Inspection requirements.

4.6.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

4.6.2 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

h. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

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i. Overheating of the engine, transmission, or any other vehicle component.

j. Evidence of corrosion.

k. Failure of the fire fighting system.

4.6.3 Detailed inspection requirements.

4.6.3.1 Examination of product. Each vehicle shall be examined to verify compliance with the requirements herein. A Government generated checklist that identifies each relevant requirement and the inspection results shall be used. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each vehicle function shall be verified. A copy of the component manufacturer's certifications shall be provided in accordance with 6.2 of NFPA 414. Prototype vehicle testing shall be in accordance with 6.3 of NFPA 414. Each production vehicle shall be inspected and operational tests conducted in accordance with 6.4 of NFPA 414. The cost associated with third party independent certification testing is the responsibility of the vehicle manufacturer.

4.6.3.1.1 Roadability test. The fully loaded first production vehicle shall be driven over ten miles of paved and ten miles of cross country terrain consisting of open fields, broken ground, and uneven terrain to demonstrate compliance with 2.1.6.1. After completing the driving portion of the roadability test, all loads shall be removed and all structure and surfaces shall be visibly inspected for failure or permanent deformation.

4.6.3.1.2 Gradeability test. The fully loaded first production vehicle shall be tested in accordance with 6.3.8 of NFPA 414 to demonstrate compliance with 2.1.6.2.

4.6.3.1.3 Tilt table test. The fully loaded first production vehicle shall be tested in accordance with 6.4.1 of NFPA 414 to demonstrate compliance with 2.1.6.3. A slip/trip rail with a maximum height of 2-inches may be used. If an adjustable height suspension system is provided, the suspension system may be set to the height normally used on hard pavement.

4.6.3.1.4 Cornering stability test. The fully loaded first production vehicle shall be tested in accordance with 6.3.2 of NFPA 414 to demonstrate compliance with 2.1.6.4.

4.6.3.1.5 Weight and weight distribution measurement. Each production vehicle shall be weighed (full and empty) to demonstrate compliance with 6.4.2 of NFPA 414. A certification shall be provided.

4.6.3.1.6 Dimension measurement. The first production vehicle shall be measured in accordance with 6.3.3 of NFPA 414. In addition to the dimensions listed in NFPA 414, the overall length, width, and height shall be measured to demonstrate compliance with 2.2.1. The height measurement shall be taken with the vehicle in its highest unloaded condition.

4.6.3.1.7 Angles of approach and departure measurement. The angles of approach and departure of the fully loaded first production vehicle shall be measured to demonstrate compliance with 2.2.3.

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4.6.3.1.8 Acceleration test. Each production vehicle shall be tested in accordance with 6.4.3 of NFPA 414 to demonstrate compliance with 2.1.6.5. For the first production vehicle, a manufacturer supplied time-distance recorder shall be used to record data for this test.

4.6.3.1.9 Top speed test. Each production vehicle shall be tested in accordance with 6.4.4 of NFPA 414 to demonstrate compliance with 2.1.6.6. For the fully loaded first production vehicle, a time-distance recorder shall be used to record data for this test.

4.6.3.1.10 Pump and roll on a 40% grade test. The fully loaded first production vehicle shall be tested in accordance with 6.3.5 of NFPA 414 to demonstrate compliance with 2.1.6.7.

4.6.3.1.11 Service / emergency brake system test. Each production vehicle shall be tested in accordance with 6.3.10 of NFPA 414. For the fully loaded first production vehicle, a time-distance recorder shall be used to record data for this test.

4.6.3.1.12 Air system and air compressor test. Each production vehicle shall be tested in accordance with 6.4.6 of NFPA 414.

4.6.3.1.13 Vehicle clearance circle test. The fully loaded first production vehicle shall be tested in accordance with 6.3.13 of NFPA 414 to demonstrate compliance with 2.3.12.2.

4.6.3.2 Cab interior sound level test. The cab interior sound levels of the first production vehicle shall be measured in accordance with 6.3.34 of NFPA 414 to demonstrate compliance with 2.4.2.

4.6.3.3 Agent system tests.

4.6.3.3.1 Agent discharge pumping test. Each production vehicle shall be tested in accordance with 6.4.7 of NFPA 414.

4.6.3.3.2 Pump and maneuver test. Each production vehicle shall be tested in accordance with 6.4.9 of NFPA 414.

4.6.3.3.3 Hydrostatic pressure test. Each production vehicle shall be tested in accordance with 6.4.10 of NFPA 414.

4.6.3.3.4 Priming device test. The priming device of the first production vehicle shall be tested in accordance with 16.10.5 of NFPA 1901.

4.6.3.3.5 Rated water and foam tank capacity test. The first production vehicle shall be tested in accordance with 6.3.1 of NFPA 414 to demonstrate compliance with the capacity requirements of 2.6.2 and 2.6.3.1.

4.6.3.3.6 Water tank fill and overflow test. The first production vehicle shall be tested in accordance with 6.3.15 of NFPA 414 to demonstrate compliance with 2.6.2.3.

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4.6.3.3.7 Foam concentration test. Each production vehicle shall be tested in accordance with 6.4.11 of NFPA 414 at the 6-percent concentration setting to demonstrate compliance with the foam concentration quality requirements of 2.6.4. This test may be conducted on-site after delivery.

4.6.3.3.8 Primary turret flow rate test. The first production vehicle shall be tested in accordance with 6.3.17 of NFPA 414. Each vehicle shall be tested to demonstrate compliance with 6.4.12 of NFPA 414. NFPA 6.4.12.4.1 shall not apply.

4.6.3.4 Electrical system tests.

4.6.3.4.1 Electrical charging system test. The first production vehicle shall be tested in accordance with 6.3.6 of NFPA 414 to demonstrate compliance with Annex B.4 of NFPA 414.

4.6.3.4.2 Electromagnetic interference test. The first production vehicle shall be tested in accordance with MIL-STD-461: CE102, CE106, CS101, CS103, CS104, CS105, CS114, CS115, CS116, RE102, and RS103 to demonstrate compliance with 2.9.4.

4.6.3.5 Ancillary equipment inventory. Each production vehicle shall be provided with a detailed inventory list of ancillary equipment. All items listed in 2.5.5 shall be inventoried and repackaged for shipment with the vehicle.

4.6.3.6 Air transportability analysis. An engineering analysis shall be performed to demonstrate compliance with the air transportability requirements of 2.2.4, 2.2.4.1, and 2.2.4.4. The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 2.2.4. The evaluation shall also include a dimensional analysis for the vehicle while traversing the ramp and while loaded aboard C-130 aircraft.

4.6.3.6.1 Equipment removal and reconfiguration demonstration. A first production vehicle shall be configured for transport on C-130 aircraft and then reconfigured for operation to demonstrate compliance with 2.2.4. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

4.6.3.6.2 Tie down provision analysis. An engineering analysis shall be performed to demonstrate compliance with the tie down provision requirements of 2.5.3.

4.6.3.6.3 Tie down provision test. A first production vehicle shall be tested to demonstrate compliance with the tie down provision requirements of 2.5.3.

5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or delivery order.

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6. NOTES.

6.1 Source of documents.

6.1.1 Department of Defense (DOD) and Federal documents may be obtained at http://quicksearch.dla.mil or from the DLA D Services, Bldg. 4D, 700 Robbins Avenue, Philadelphia PA 19111-5094.

6.1.2 The Code of Federal Regulations (CFR) may be obtained at http://www.gpo.gov/fdsys/search/home.action or from the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402.

6.1.3 Society of Automotive Engineers (SAE) documents may be obtained at http://www.sae.org/servlets/index or from SAE, Inc., 400 Commonwealth Drive, Warrendale PA 15096.

6.1.4 National Fire Protection Association (NFPA) documents may be obtained at http://www.nfpa.org/index.asp or from NFPA, Batterymarch Park, Quincy MA 02269-9101.

6.1.5 FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from http://www.arnet.gov/far Electronic copies of the DFARS may be obtained from http://www.acq.osd.mil/dpap/dars/dfars/index.htm.

6.1.6 Occupational Safety and Health Association (OSHA) copies may be obtained online at http://www.osha.gov/ or from OSHA, 200 Constitution Ave., Washington, D.C. 20210.

6.1.7 Environmental Protection Agency (EPA) documents may be obtained at http://www.epa.gov/ or from USEPA Ariel Rios Building (AR), 1200 Pennsylvania Avenue N.S., Washington, DC 20004.

6.2 Ordering Data. The contract or order should specify the following:

a. Emergency warning light color required (amber or blue in place of red) (see 2.9.7.1).

b. Electrical shoreline voltage, 220 volts instead of 110 volts (see 2.9.3.1 and 2.3.11.4).

6.3 Key acronyms.

GPM	Gallons Per Minute
PSI	Pounds Per Square Inch
NHT	National Hose Thread

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